Notice the *i* in the second definition. Now, substitute 5.10 into 5.9 to get

$$2\sqrt{\langle \mathbf{A}^2\rangle\langle \mathbf{B}^2\rangle} \geq |\langle\Psi|\mathbf{AB}|\Psi\rangle - \langle\Psi|\mathbf{BA}|\Psi\rangle|. \qquad (5.11)$$

The minus sign is due to the factor of *i* in the second definition in 5.10. Using the definition of a commutator, we find that

$$2\sqrt{\langle \mathbf{A}^2\rangle\langle \mathbf{B}^2\rangle} \geq |\langle\Psi|[\mathbf{A},\mathbf{B}]|\Psi\rangle|. \qquad (5.12)$$

Let's suppose for the moment that **A** and **B** have expectation values of zero. In that case, $<\mathbf{A}^2>$ is just the square of the uncertainty in A, that is, $(\Delta\mathbf{A})^2$, and $<\mathbf{B}^2>$ is just $(\Delta\mathbf{B})^2$. Thus we can rewrite Eq. 5.12 as

$$\Delta\mathbf{A}\ \Delta\mathbf{B} \geq \frac{1}{2}|\langle\Psi|[\mathbf{A},\mathbf{B}]|\Psi\rangle|. \qquad (5.13)$$

Reflect on this mathematical inequality for a moment. On the left side, we see the product of the uncertainties of the two observables **A** and **B** in the state Ψ. The inequality says that this product cannot be smaller than the right side, which involves the commutator of **A** and **B**. Specifically, it says that the product of the uncertainties cannot be smaller than *half the magnitude of the expectation value of the commutator*.

The general uncertainty principle is a quantitative expression of something we already suspected: if the commutator of **A** and **B** is not zero, then both observables cannot simultaneously be certain.

But what if the expectation value of **A** or **B** is not zero? In that case, the trick is to redefine two new operators in which the expectation values have been subtracted off: